

PRATEEK SAHU

PhD Candidate, Computer Architecture
University of Texas at Austin

prateeks@utexas.edu
<https://www.linkedin.com/in/sahuprateek/>
<https://prateeksahu.github.io>

SUMMARY

Defensive security researcher specializing in cross-stack security research — from hardware microarchitectural side-channels and runtime ransomware detection to vulnerabilities in modern AI inference pipelines. Published at MICRO and ASPLOS, with ongoing work on AI threat red-teaming and adversarial attack amplification across the software-hardware boundary. Seeking research roles at the intersection of offensive security and frontier AI safety.

Research Interests: AI Security, Systems Security, Malware Detection, Microarchitectural Side Channels

EDUCATION

Ph.D., Computer Architecture • *University of Texas at Austin* 2020 - Present (Expected June 2026)

Advisor: Prof. Mohit Tiwari

Dissertation: Identifying and Detecting Cross Layer Threats in Modern Workloads

Qualcom Innovation Fellowship recipient (2022)

M.Sc., Computer Architecture • *University of Texas at Austin* 2017 - 2019

GPA: 3.79/4

Bachelor of Engineering, Electrical • *Indian Institute of Technology, Kanpur* 2011 - 2015

GPA: 8.2/10

Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship, Dept. of Science and Technology, India

RESEARCH EXPERIENCE

Graduate Research Assistant • *SPARK Lab, UT Austin* • *Prof. Mohit Tiwari* Spring 2018 - Present

AI Systems Security Research (2023 - Present)

- Systematization of hardware and software threat vectors in modern compound AI inference pipelines (arXiv '24)
- Developed Cascade – AI red-team framework demonstrating how software-hardware attack gadgets compose to amplify adversarial threats in AI systems; exemplified using fault injection and model jailbreak (arXiv '26)
- Investigated prompt injection vulnerabilities in M365 production deployments to suppress correct responses and misattribution of ownership using malicious RAG injection (arXiv '24)

Ransomware Detection Research (2021 - 2023)

- Designed a lifecycle-aware ransomware detector leveraging cross-layer signals – hardware performance counters, OS events, and network telemetry – for early-stage detection with reduced data loss
- Trained detection model on MITRE ATT&CK malware lifecycle stages; demonstrated higher detection confidence vs. single-layer approaches

Micro-architectural Side Channel Research (2018 - 2021)

- Developed speculative interference attacks that exploit younger speculative instructions affecting older instruction latency, undermining existing cache defense mechanisms (ASPLOS '21).
- Created Cyclone, a side-channel malware detector that tracks cyclic contention events across security domains, evaluated against cache-based side and covert channels on the gem5 simulation environment (MICRO '19).

WORK EXPERIENCE

Graduate Technical Intern • *Intel Labs* Jun - Nov 2020, May - Aug 2021

- Built proof-of-concept for secure data offload pipeline in Intel SGX-adjacent accelerator environments.
- Designed remote attestation and key provisioning flows for confidential compute offload use cases.

Engineering Intern • *Qualcomm Technologies* May - Aug 2019

- Architectural design for SMT in QDSP6 Control Unit and their security implications.
- Formal verification of existing RTL design to find hardware scheduler bugs.

Member of Technical Staff • *VMware India* July 2015 - June 2017

- Filed 3 patents with USPTO for a system health monitoring tool using vSphere metrics.
- Designed and implemented a private cloud resource and cost monitoring tool using utilization statistics.
- Developed containerized micro-services CI/CD pipelines for vSphere application infrastructure.

SELECTED PUBLICATIONS

- Cascade: Composing Software-Hardware Attack Gadgets for Adversarial Threat Amplification in Compound AI Systems (arXiv '26)
Sahu, P.*, Banerjee, S.*, Vahldiek-Oberwagner, A., Sanchez, J.V., Tiwari, M. [[pdf](#)]
- SoK: A systems perspective on compound AI threats and countermeasures (arXiv '24)
Sahu, P.*, Banerjee, S.*, Luo, M., Vahldiek-Oberwagner, A., Yadwadkar, N.J., Tiwari, M. [[pdf](#)]
- Understanding Sidecars in Cloud Orchestration (SESAME '25, co-located with EuroSys '25)
Sahu, P., Wei, S., Yadwadkar, N.J., Tiwari, M. [[pdf](#)]
- Speculative Interference Attacks: Breaking Invisible Speculation Schemes (ASPLOS '21)
Behnia, M., **Sahu, P.**, Paccagnella, R., Yu, J., Zhao, Z., Zou, X., Unterluggauer, T., Torrellas, J., Rozas, C., Morrison, A., Mckeen, F., Liu, F., Gabor, R., Fletcher, C.W., Basak, A., Alameldeen, A. [[pdf](#)]
- Cyclone: Detecting Contention-Based Cache Information Leaks Through Cyclic Interference (MICRO '19) Harris, A., Wei, S., **Sahu, P.**, Kumar, P., Austin, T., Tiwari, M. [[pdf](#)]

TECHNICAL SKILLS

AI/ML: PyTorch, LLM inference pipelines, RAG systems, agentic frameworks

Languages: C, C++, Python, Bash, x86 Assembly

Simulation: gem5, QEMU

System Tools: Perf, eBPF, Intel PT, Valgrind